

Effect of the type of coagulant on coagulation properties of rubbers, compounds and vulcanisates

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Polymers produced using an emulsion (co)polymerisation method are one of the most important products in world industry. The main advantages offered by this method are the absence of organic solvents with their attendant fire hazard and threat to human health, and the relatively high physical and mechanical values of rubbers and vulcanisates based on them. As a result, emulsion rubbers have become widely used in the tyre and industrial rubber products industry [1-5].

Emulsion rubbers are currently produced from latexes using an aqueous sodium chloride solution with subsequent acidification using sulphuric acid. The main drawback to this method is the high consumption of coagulant, which is 180-350 kg per tonne of rubber. The aqueous solution containing sodium chloride, sulphuric acid and other water-soluble components (serum) is returned in part for coagulation, while the majority is sent to a purification plant. Sodium chloride, which is highly stable, does not break down at the purification plant and gets into natural water bodies, leading to soil and ground water contamination. The search for new coagulants possessing a high coagulation capacity and capable of helping to produce rubber from latex with a low consumption is consequently an important and urgent task. Good results have been obtained using ammonium salts to produce rubber from latex [6-10]. The high cost of these reagents, however, is holding back their commercial-scale introduction.

Mention has been made of other coagulants for the production of emulsion rubber from latex [11-13] – salt

meal and lignin and its derivatives. But there are certain difficulties in using these compounds under industrial conditions.

In the 1980s, protein coagulants were used on a commercial scale [14,15], enabling use of mineral salts to be discontinued. Aqueous solutions of protein coagulants are relatively unstable, however, especially at high temperatures. Breakdown of proteins leads to gases being released into production premises.

A study has also been made [16] of the potential to replace sodium chloride with special coagulants (VMS-100, VMS-100A) which are non-toxic and break down readily during biological purification. The physical and mechanical values for rubber obtained using the salt-free process are similar to those obtained for rubber types produced from latex using sodium chloride. The rubber obtained features a high level of property stability. These coagulants do, however, have a number of serious drawbacks: a limited storage period, instability and high cost.

From the information available in the specialist literature, and from experience gained at rubber production plants, the most promising coagulants are polyvalent metal salts. Salt consumption is substantially reduced when these are employed [17]. The available information is, however, behind the times, as the synthetic polymer industry, and especially emulsion rubber manufacture, nowadays employs different emulsifying systems. Such emulsifiers as *nekal* (according to the Alford & Alford Dictionary, *nekal* is "sodium dibutyl-naphthalene sulphonate" – Translator)

and *kontakt Petrova* (according to Callaham's Dictionary, 4th ed., *kontakt Petrova* is "a detergent mixture of sulfonaphthenic acids" – *Translator*) have been excluded from the process as biologically non-degradable. The replacement of harmful and toxic emulsifiers by substances that are less toxic and less resistant to biological degradation has duly also led to a change in the stability of colloid systems (latexes).

Widespread use is currently being made on a commercial scale of emulsifying agents based on tall oil, disproportionated rosin and fatty acid salts.

The aim of this work was to look at the process of butadiene-styrene rubber production from SKS-30ARK latex (dry residue 18%) using chlorides of sodium, calcium, aluminium and tin(IV) as coagulation agents.

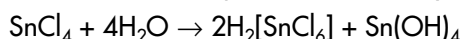
For production of rubber from latex, 20 cm³ of latex was loaded into a coagulator, this was thermostatted at a preset temperature for 15-20 minutes, then aqueous solutions of the coagulation agents and sulphuric acid were introduced; the pH was kept constant (~ 2.0) by introducing a 1-2% aqueous sulphuric acid solution.

The aqueous solution concentrations were:

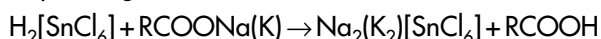
NaCl – 24%, CaCl₂, AlCl₃·6H₂O and SnCl₄·5H₂O – 10%.

A study was made of the influence of temperature, consumption level and nature of the coagulation agents on the completeness of rubber production from latex (**Figure 1, Table 1**).

It was demonstrated that increasing the metal cation charge from +1 to +3 helps to reduce coagulant consumption; this correlates well with the results of works published previously. The consumption of tin(IV) chloride is greater than the consumption of aluminium chloride needed for full rubber production from latex, however, and in contrast to AlCl₃, it increases as the process temperature rises. One possible reason why tin(IV) chloride behaves like this may be hydrolysis of it in aqueous solution to form a complex acid and tin hydroxide [18]:



The acid generated is a strong acid and may react with surfactants in the emulsifier system, giving rise to the corresponding salts:



The salts formed are resistant to hydrolysis. The acid medium created in an aqueous tin chloride solution enables acidification of the coagulation system by sulphuric acid to be not just scaled back, but eliminated altogether.

Study of rubber mixtures and vulcanisates based on SKS-30ARK rubber revealed that the nature of the coagulation agent does not have any major impact on their properties. Rubber mixtures based on rubber produced from latex using tin chloride, however, feature a higher Karrer plasticity and hardness and lower rebound and resiliency (**Table 2**).

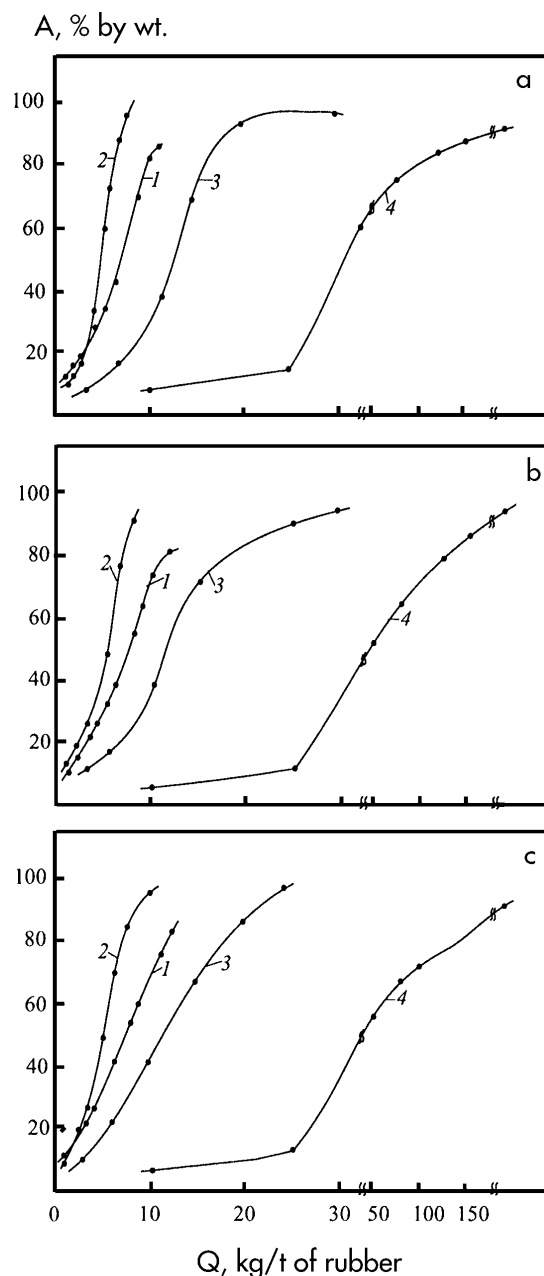


Figure 1. Graph of fullness of production (A) of rubber from SKS-30ARK latex at a temperature of (a) 40, (b) 60 and (c) 80 °C against consumption of SnCl₄ (1), AlCl₃ (2), CaCl₂ (3) and NaCl (4).

Table 1. Impact of the nature of the coagulation agent in production of SKS-30ARK rubber from latex on its consumption			
Coagulation agent	Coagulation agent consumption, kg/t of rubber, at temperature		
	40 °C	60 °C	80 °C
NaCl	250	250	250
CaCl ₂	25	30	30
AlCl ₃	9	9	9
SnCl ₄	11	12	12

The vulcanisation rate for rubber mixtures based on SKS-30ARK rubber depends on the nature of the coagulation agent used (**Table 3**), and diminishes in the following sequence: $\text{CaCl}_2 > \text{SnCl}_4 > \text{AlCl}_3 > \text{NaCl}$.

The results obtained permit the conclusion that the most promising coagulation agents are those based on aluminium salts. Aluminium chloride consumption is not affected by the temperature regimen for the process of rubber production from latex. The nature of the coagulation agent does not have any major impact on the properties of the rubber, rubber mixtures or vulcanisate obtained.

Table 2. Properties of rubber mixtures and vulcanisates based on SKS-30ARK rubber				
Indicator	Coagulation agent			
	NaCl	CaCl_2	AlCl_3	SnCl_4
Mooney plasticity (MB 1+4, 100 °C), units	45.0	50.0	49.0	46.0
Karrer plasticity, units	0.39	0.37	0.37	0.42
Rebound, mm	1.46	1.69	1.65	1.38
Rel. stress at 300% elongation, MPa	<u>9.2</u>	<u>8.3</u>	<u>8.7</u>	<u>10.3</u>
	11.1	8.9	9.8	12.0
Rel. tensile strength, MPa	<u>24.7</u>	<u>25.1</u>	<u>25.0</u>	<u>26.2</u>
	24.5	24.4	25.2	26.4
Ultimate elongation, %	<u>580</u>	<u>610</u>	<u>600</u>	<u>560</u>
	510	570	560	500
Rel. permanent set following rupture, %	<u>12</u>	<u>20</u>	<u>14</u>	<u>14</u>
	10	16	14	12
Resiliency, %	42	40	40.5	39
Shore A hardness, units	57	59	9	61
<i>Note: Vulcanisation temperature 143 °C; duration of vulcanisation 60 min (numerator) and 80 min (denominator)</i>				

Table 3. Characteristics of vulcanisation process for rubber mixtures based on SKS-30ARK rubber (at 100 °C)				
Indicator	Coagulation agent			
	NaCl	CaCl_2	AlCl_3	SnCl_4
Minimum torque, N·m	4.8	4.6	5.1	5.1
Maximum torque, N·m	36.5	33.0	33.5	36.2
Vulcanisation start time, min	3.5	4.0	4.0	4.0
Time when 50% vulcanisation level achieved, min	10.2	10.7	11.5	9.9
Optimum vulcanisation time, min	27.5	18.5	26.0	23.0
Rate of vulcanisation, min ⁻¹	4.17	6.9	4.54	5.26

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